



FAKULTI SAINS KOMPUTER DAN MATEMATIK (FSKM)

SEMESTER II 2025/2026

CSE3433 SOFTWARE ARCHITECTURE (K2)

Individual Log Book

GROUP 8

Service-Oriented Smart Campus Support System

(Service-Oriented Architecture)

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PHASE 1 : Project Proposal (Week 3 - Week 5)

Week	Activities Undertaken	Individual Contribution	Challenges	Action Plan
3	Initial system idea & concept formation.	Researched SOA concepts and contributed ideas during discussions to define the Smart Campus Support System concept and identify suitable services.	Unclear system scope and SOA architecture direction.	Conduct deeper research on SOA implementation and study similar service-based systems. Identify core services required for the project and validate system feasibility before proceeding to the requirement analysis phase.
4	Requirement analysis & project validation .	Analysed project requirements, refined system scope, and assisted in identifying user roles and service functionalities.	Difficulty narrowing system scope to 4 services .	Evaluate system functionalities and prioritise essential services only. Separate mandatory and optional features to ensure the system remains manageable while maintaining compliance with SOA principles.
5	Proposal drafting & finalisation .	Prepared proposal content, organised requirements documentation, and reviewed project scope before submission.	Time pressure before submission deadline.	Allocate dedicated review sessions to validate proposal content. Perform proofreading, formatting corrections, and consistency checks to improve overall document quality and submission readiness.

PHASE 2: Architecture Design (Week 6 - Week 7)

Week	Activities	Individual	Challenges	Action Plan
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	Undertaken	Contribution		
6	Study SOA & REST API architecture.	Studied SOA principles and REST API communication to understand service interaction and architecture implementation.	Limited understanding of SOA principles and service communication	Doing extra research and reviewing additional technical references and architecture examples. Create simplified service interaction diagrams to visualise request flows and improve understanding of SOA concepts.
7	Architecture design and system diagram preparation.	Assisted in designing system architecture diagrams and database structures while ensuring alignment with project requirements.	Complexity in diagram structuring for service routing	Refine architecture diagrams through multiple revisions and validate relationships between services. Ensure consistency between architecture diagrams, database structures, and system workflows.

PHASE 3: Prototype progress (Week 8 - Week 11)

Week	Activities Undertaken	Individual Contribution	Challenges	Action Plan
8	Prototype development and initial testing.	Defined servlet responsibilities, reviewed system workflow, and participated in initial prototype testing activities.	UI inconsistency in early JSP prototype.	Establish a consistent user interface design standard and review all JSP pages for uniformity. Improve navigation structure and gather feedback for interface

				refinement.
9	System evaluation & debugging.	Tested database connectivity and service communication, identified integration issues, and documented testing outcomes.	Database and HttpURLConnection integration issues	Test services individually to isolate errors and identify root causes. Improve documentation of service interactions and refine integration procedures to minimise communication failures.
10	Frontend, backend integration and deployment preparation	Developed JSP pages, configured Tomcat deployment settings, and performed deployment testing.	First-time deployment experience using Apache Tomcat 9	Follow deployment documentation carefully and conduct multiple deployment trials. Record configuration settings and troubleshooting steps to facilitate future deployments.
11	System enhancement, testing and deployment	Improved user interface design, fixed system issues, and conducted testing to ensure system stability.	Deployment configuration and servlet mapping issues	Verify servlet mappings and deployment settings systematically. Perform end-to-end testing and continuously monitor application behaviour to identify remaining issues.

PHASE 4: Project Report, Individual Logbook & Final Demo (Week 12 - Week 13)

Week	Activities Undertaken	Individual Contribution	Challenges	Action Plan
12	Final system review and report preparation	Reviewed system functionality, verified project requirements, and assisted in report preparation activities.	Documentation structure and content organisation.	Review report structure against the official template and reorganise content where necessary. Validate technical descriptions and ensure consistency across all documentation sections.
13	Final report submission, presentation, and system demo	Finalised project documentation, prepared presentation materials, and participated in system demonstration preparation.	Formatting inconsistencies in documentation and nervousness for presentation.	Perform final proofreading and formatting reviews across all deliverables. Conduct multiple presentation rehearsals and prepare responses for potential technical questions during evaluation.

GEN AI USAGE RECORD

Week	Date	Tool Used	Purpose	Prompt Used	Output Summary	How It Was Used
Week 3	28/03/2026	Gemini	Architecture orientation	"explain fundamental concepts of Service-Oriented Architecture for university assignment"	Provided a clear breakdown of loose coupling, service reusability, and standard protocols.	Used to build baseline understanding during early team conceptualization.

Week 3	30/03/2026	Gemini	Scope formulation	"give me ideas for a campus management system using 4 specific services under SOA"	Suggested dividing the system into Helpdesk, Academic, Notification, and Inventory services.	Used to evaluate alternative service boundaries before finalizing the ticketing focus.
Week 4	05/04/2026	Gemini	Problem identification	"how to draft a strong problem statement for disjointed legacy systems in a university"	Outlined major pain points like data siloing, manual routing delays, and lack of notifications.	Integrated into the background and problem statement sections of the proposal.
Week 4	08/04/2026	Gemini	Scope refinement	"what are typical user journeys in an automated campus ticketing platform"	Map out step-by-step interactions for student submitters, IT staff, and system admins.	Used to define clear system boundaries and avoid feature creep.
Week 5	12/04/2026	Gemini	Requirements gathering	"list non-functional requirements vital for a web-based SOA project"	Highlighted essential parameters like response time, horizontal scalability, and API authentication	Added to the system specification documentation in the proposal.

					safety.	
Week 5	15/04/2026	Gemini	Proposal polishing	"review this structure of a software architecture proposal for clarity and flow"	Provided structural recommendations to improve the methodology and objective alignment.	Used to refine the draft before the official Phase 1 submission deadline.
Week 6	21/04/2026	Gemini	Service decomposition	"how does an API Gateway or Router Servlet direct traffic to independent department servlets"	Explained the mechanism of reading request paths and forwarding payloads via HTTP.	Guided the design of the centralized RouterServlet workflow.
Week 7	26/04/2026	Gemini	Database planning	"how to structure separate databases for IT and Facility departments to keep them decoupled"	Provided schema examples with isolated primary keys and cross-service reference strategies.	Applied directly to the architectural data modeling phase (db design).
Week 7	29/04/2026	Gemini	Protocol validation	"verify if using HttpURLConnection for internal servlet communication"	Confirmed it fulfills loose coupling and statelessness via RESTful principles.	Used as technical justification in the architectural design documentation.

				on adheres to SOA"		on.
Week 8	10/05/2026	Gemini	Initial prototyping	"minimalist boilerplate for a Java HttpServlet handling JSON POST requests"	Supplied a clean template utilizing doPost with BufferedRead er handling.	Used to set up the baseline code structure in NetBeans.
Week 9	17/05/2026	Gemini	API Integration	"how to parse JSON responses from an internal endpoint using vanilla Java"	Detailed string tokenization and lightweight JSON parsing techniques suitable for servlets.	Used to establish seamless communication between the router and sub-services.
Week 9	19/05/2026	Gemini	Testing strategies	"create a Postman collection test strategy for an asynchronous notification endpoint"	Provided a walkthrough on setting up environment variables and verifying HTTP 202/200 statuses.	Used to debug and validate servlet endpoints systematically.
Week 10	24/05/2026	Gemini	Server deployment	"step by step guide to map web.xml for multiple servlets in Apache	Explained context paths, servlet-name bindings, and URL patterns clearly.	Used to resolve initial deployment configuration n hitches.

				Tomcat 9"		
Week 10	27/05/2026	Gemini	Troubleshooting	"how to fix HTTP Status 404 Not Found error on Tomcat when servlet mapping seems correct"	Diagnosed common compilation target mismatches and deployment directory errors.	Used to resolve localized server mapping and configuration bugs.
Week 11	02/06/2026	Gemini	Interface enhancement	"modern and clean CSS styling ideas for an internal ticketing submission form"	Provided a responsive, layout-focused CSS snippet using a soft, professional color palette.	Used to elevate the look of index.jsp and results pages.
Week 12	09/06/2026	Gemini	Documentation	"how to format an architecture evaluation report comprehensively"	Suggested a structure containing system stress test results, metric tables, and failure analysis.	Used to compile the technical report deliverables.
Week 13	16/06/2026	Gemini	Final validation	"checklist to ensure a Java-based SOA project is ready for production demo"	Provided a comprehensive list covering exception handling, state verification, and UI	Used as a final QA assessment before submission.

					responsiveness.	
Week 13	18/06/2026	Gemini	Viva preparation	"what technical questions do examiners usually ask about RESTful servlets and loose coupling"	Listed questions regarding state management, error propagation, and database independence.	Used to script the final presentation and rehearse defensive answers.